

Test Form E

Name _____ Date _____

Chapter 11

Class _____ Section _____

1. A vector $\mathbf{v}$ has initial point $(2, -1)$ and terminal point $(-2, 3)$. Find the unit vector in the direction of $\mathbf{v}$.
2. Find a vector with magnitude 3 in the direction of the vector $\mathbf{v} = \langle 1, 2 \rangle$.
3. Three forces with magnitude of 50, 20, and 40 pounds act on an object at angles of 60° , 30° , and -90° , respectively, with the positive x -axis. Find the direction and magnitude of the resultant forces.
4. Find the standard equation for the sphere that has points $(-2, 5, -1)$ and $(3, 7, -3)$ as endpoints of a diameter.
5. Determine whether each vector is parallel to the vector $\mathbf{v} = -4\mathbf{i} - \mathbf{j} + 5\mathbf{k}$. If it is, find c such that $\mathbf{v} = c\mathbf{u}$.
 - a. $\mathbf{u} = 8\mathbf{i} + 4\mathbf{j} + 10\mathbf{k}$
 - b. $\mathbf{u} = -8\mathbf{i} - 4\mathbf{j} + 10\mathbf{k}$
 - c. $\mathbf{u} = 3\mathbf{i} + \frac{3}{4}\mathbf{j} - \frac{15}{4}\mathbf{k}$
 - d. $\mathbf{u} = -2\mathbf{i} + \frac{1}{2}\mathbf{j} + \frac{5}{2}\mathbf{k}$
6. A vector $\mathbf{v}$ has initial point $(2, 1, 3)$ and terminal point $(-4, 2, -1)$.
 - a. Write $\mathbf{v}$ in the component form.
 - b. Write $\mathbf{v}$ as a linear combination of the standard unit vectors.
 - c. Find the magnitude of $\mathbf{v}$.
 - d. Find the unit vector in the direction of $\mathbf{v}$.
 - e. Find the unit vector in the direction opposite that of $\mathbf{v}$.
7. Find $\mathbf{u} \cdot \mathbf{v}$ if $\|\mathbf{u}\| = 7$, $\|\mathbf{v}\| = 12$, and the angle between $\mathbf{u}$ and $\mathbf{v}$ is $\frac{\pi}{4}$.
8. Let $\mathbf{u} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$.
 - (a) Find the projection of $\mathbf{u}$ onto $\mathbf{v}$.
 - (b) Find the vector component of $\mathbf{u}$ orthogonal to $\mathbf{v}$.
9. Find direction cosines for the vector with initial point $(2, -1, 3)$ and terminal point $(4, 3, -5)$.
10. Determine a scalar, k , so that the vectors $\mathbf{u} = 3\mathbf{i} + 2\mathbf{j}$ and $\mathbf{v} = 2\mathbf{i} + k\mathbf{j}$ are orthogonal.

11. Let $\mathbf{u} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$.
- Find $\mathbf{u} \times \mathbf{v}$.
 - Show that the vector $\mathbf{u} \times \mathbf{v}$ is orthogonal to $\mathbf{v}$.
12. Calculate the area of the triangle with vertices $(3, -1, 2)$, $(2, 1, 5)$, and $(1, -2, -2)$.
13. Find parametric equations for the line perpendicular to the lines given by $\frac{x-2}{-4} = \frac{y+3}{-7} = \frac{z+1}{3}$ and $\frac{x-3}{7} = \frac{y+2}{-2} = \frac{z-8}{3}$ passing through the point $(2, 4, -1)$.
14. Find the point of intersection of the line $\frac{x+2}{2} = \frac{y-3}{3} = \frac{z-1}{1}$ and the plane $2x + 3y + z - 3 = 0$.
15. Sketch the surface represented by $y = x^2 + z^2$.
16. Identify each of the quadric surfaces:
- $3x^2 + 3y^2 + 3z^2 - 2x + 3y - 11 = 0$
 - $2x^2 + 5z^2 = y^2$
 - $2x^2 - y^2 - z^2 = 1$
17. Consider the surface whose rectangular equation is $x^2 + y^2 - z^2 = 1$.
- Find an equation for the surface in cylindrical coordinates.
 - Find an equation for the surface in spherical coordinates.
18. Sketch the surface represented by the spherical equation $\rho = 2 \csc \phi$.